

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 6: LECTURE NOTE

POLYNOMIAL FUNCTIONS

Definition 1. A polynomial function is a function of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where n is a nonnegative integer and a_j , $j = 0, 1, 2, \dots, n$ are real numbers with $a_n \neq 0$. The number a_n is called the leading coefficient and a_0 its constant term. Such a polynomial is called a polynomial of degree n .

We observe that

- The domain of a polynomial function is the set of all real numbers, $\mathbb{R} = (-\infty, \infty)$.
- Polynomials of degree 0, $f(x) = a_0$, are called **constant functions**. The graph of a constant function is a **horizontal line**.
- Polynomials of degree 1, $f(x) = ax + b$, $a \neq 0$, are called **linear functions**. The graph of a linear function is a **line** whose nature depends on the sign of a .
- Polynomials of degree 2, $f(x) = ax^2 + bx + c$, $a \neq 0$, are called **quadratic functions**. The graph of a quadratic function is either an upward ($a > 0$) or a downward ($a < 0$) **parabola**.
- A polynomial of degree 3 is called a cubic polynomial,

$$f(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0, \quad a_3 \neq 0.$$

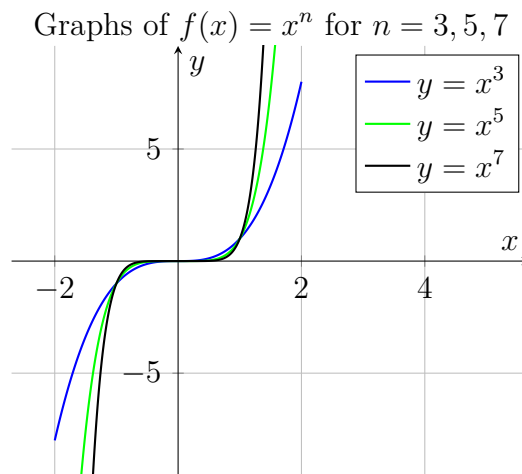
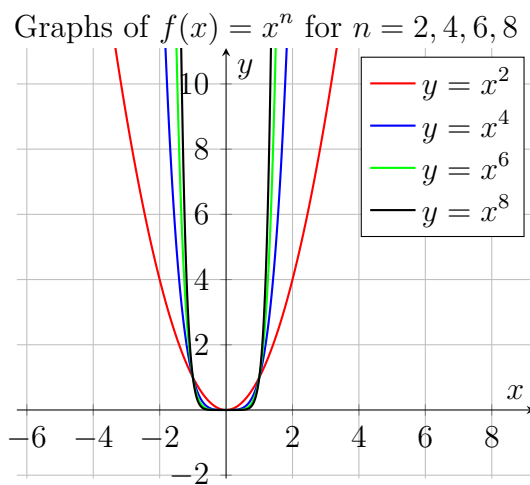
Features of the graph of a polynomial function

1. A polynomial function has a **continuous graph**. The graph of a polynomial function has **no breaks, holes, or gaps** (Continuous). We say the graph of a function is continuous when its graph can be drawn with a pencil without lifting the pencil from the paper.
2. The higher degree polynomial functions that have the simplest graphs are monomials of the form

$$f(x) = x^n$$

where n is an integer greater than zero. Polynomial functions of the form $f(x) = x^n$ are often referred to as power functions.

- (a) The greater the value of n , the flatter the graph near the origin.
- (b) When n is even, the graph is similar to the graph of $f(x) = x^2$ and touches the x -axis at the x -intercept.
- (c) When n is odd, the graph is similar to the graph of $f(x) = x^3$ and crosses the x -axis at the x -intercept.



3. The graph of a polynomial function has only **smooth, rounded turns**. It cannot have a **sharp turn**.
4. The graph of a polynomial function of degree n can have at most n zeros; that is, the polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0 = 0$$

has at most n solutions.

5. The graph of a polynomial function always exhibits the characteristic that as $|x|$ gets very large $|f(x)|$ also gets very large.

LONG DIVISION OF POLYNOMIAL FUNCTIONS

Theorem 2 (DIVISION ALGORITHM). *If $f(x)$ and $d(x)$ are polynomial such that $d(x) \neq 0$, and the degree of $d(x)$ is less than or equal to the degree of $f(x)$, then there exists unique polynomials $q(x)$ (called quotient polynomial) and $r(x)$ (called the remainder polynomial) such that*

$$f(x) = d(x).q(x) + r(x)$$

where $r(x) = 0$ or the degree of $r(x)$ is less than the degree of $d(x)$. If the remainder $r(x)$ is zero, then $d(x)$ divides evenly into $f(x)$.

Example 3 (LONG DIVISION). *Consider the polynomial functions*

$$f(x) = 6x^3 - 19x^2 + 16x - 4 \quad \text{and} \quad d(x) = x - 2.$$

Because

$$f(2) = 6(2)^3 - 19(2)^2 + 16(2) - 4 = 0$$

we know that $x = 2$ is a solution of the polynomial function f , and that $(x - 2)$ is a factor of f . This means that there exists a second degree polynomial $g(x)$ such that

$$f(x) = (x - 2).q(x).$$

To find $q(x)$, we can use **long division of polynomial**. Divide $6x^3 - 19x^2 + 16x - 4$ by $x - 2$, and use the result to factor the polynomial completely.

$$\begin{array}{r} \quad \quad \quad 6x^2 - 7x + 2 \\ x-2) \overline{ 6x^3 - 19x^2 + 16x - 4} \\ \underline{- 6x^3 + 12x^2} \\ - 7x^2 + 16x \\ \underline{7x^2 - 14x} \\ 2x - 4 \\ \underline{- 2x + 4} \\ 0 \end{array}$$

You can see that

$$\frac{6x^3 - 19x^2 + 16x - 4}{x - 2} = 6x^2 - 7x + 2$$

or

$$\begin{aligned} 6x^3 - 19x^2 + 16x - 4 &= (x - 2)(6x^2 - 7x + 2) \\ &= (x - 2)(2x - 1)(3x - 2) \end{aligned}$$

$x - 2$ is a factor of the polynomial $6x^3 - 19x^2 + 16x - 4$ and the long division process produces a remainder of zero.

Example 4. *LONG DIVISION*

1. Divide $f(x) = x^2 + 3x + 5$ by $d(x) = x + 1$.

$$\begin{array}{r} x+2 \\ x+1 \overline{) x^2+3x+5} \\ \underline{-x^2 \quad -x} \\ 2x+5 \\ \underline{-2x-2} \\ 3 \end{array}$$

Here $q(x) = x + 2$ and $r(x) = 3$. In fractional form, you can write this result as follows

$$\frac{x^2 + 3x + 5}{x + 1} = x + 2 + \frac{3}{x + 1}.$$

This implies that

$$f(x) = x^2 + 3x + 5 = (x + 2)(x + 1) + 3.$$

2. Divide $f(x) = 8x^3 - 1$ by $d(x) = 2x - 1$.

$$\begin{array}{r}
 4x^2 + 2x + 1 \\
 2x - 1 \overline{) 8x^3 - 1} \\
 \underline{- 8x^3 + 4x^2} \\
 4x^2 \\
 \underline{- 4x^2 + 2x} \\
 2x - 1 \\
 \underline{- 2x + 1} \\
 0
 \end{array}$$

Here the quotient is $q(x) = (4x^2 + 2x + 1)$ and the remainder is zero. So, $d(x) = 2x - 1$ divide evenly into $8x^3 - 1$, and we can write

$$f(x) = 8x^3 - 1 = (2x - 1)(4x^2 + 2x + 1) = d(x)q(x).$$

3. Divide $-2 + 3x - 5x^2 + 4x^3 + 2x^4$ by $x^2 + 2x - 3$.

$$\begin{array}{r}
 2x^2 + 1 \\
 x^2 + 2x - 3 \overline{) 2x^4 + 4x^3 - 5x^2 + 3x - 2} \\
 \underline{- 2x^4 - 4x^3 + 6x^2} \\
 x^2 + 3x - 2 \\
 \underline{- x^2 - 2x + 3} \\
 x + 1
 \end{array}$$

Here $q(x) = 2x^2 + 1$ and $r(x) = x + 1$.

4. Divide $f(x) = x^4 - 1$ by $d(x) = x^2 + 2x$.

$$\begin{array}{r}
 x^2 - 2x + 4 \\
 x^2 + 2x \overline{) x^4 - 1} \\
 \underline{- x^4 - 2x^3} \\
 - 2x^3 \\
 \underline{2x^3 + 4x^2} \\
 4x^2 \\
 \underline{- 4x^2 - 8x} \\
 - 8x
 \end{array}$$

Here $q(x) = x^2 - 2x + 4$ and $r(x) = -8x$.

5. Divide $f(x) = x^4 + x^3 + 2x^2 + 3x + 1$ by $d(x) = x^2 + 4$.

$$\begin{array}{r}
 x^2 + x - 2 \\
 x^2 + 4 \overline{) x^4 + x^3 + 2x^2 + 3x + 1} \\
 \underline{- x^4 - 4x^2} \\
 x^3 - 2x^2 + 3x \\
 \underline{- x^3 - 4x} \\
 - 2x^2 - x + 1 \\
 \underline{2x^2 + 8} \\
 - x + 9
 \end{array}$$

Here $q(x) = x^2 + x - 2$ and $r(x) = -x + 9$.

THE REMAINDER AND FACTOR THEOREM

Theorem 5 (THE REMAINDER THEOREM). *If a polynomial $f(x)$ of degree $n \geq 1$ is divided by $d(x) = x - a$, then the remainder is*

$$r(x) = f(a).$$

That is

$$f(x) = d(x)q(x) + f(a).$$

Example 6. *Find the remainder when $f(x) = 3x^3 + 8x^2 + 5x - 7$ is divided by $x + 2$. By the remainder theorem, the remainder is*

$$f(-2) = 3(-2)^3 + 8(-2)^2 + 5(-2) - 7 = -9.$$

The LONG DIVISION also gives the same:

$$\begin{array}{r} \overline{3x^2+2x+1} \\ x+2 \overline{) 3x^3+8x^2+5x-7} \\ \underline{-3x^3-6x^2} \\ 2x^2+5x \\ \underline{-2x^2-4x} \\ x-7 \\ \underline{-x-2} \\ -9 \end{array}$$

Here $q(x) = 3x^2 + 2x + 1$ and $r(x) = -9$.

Theorem 7 (THE FACTOR THEOREM). *A polynomial $f(x)$ has a factor $x - a$ if and only if $f(a) = 0$.*

Example 8. (THE FACTOR THEOREM)

1. $x - 2$ and $x + 3$ are factors of

$$f(x) = 2x^4 + 7x^3 - 4x^2 - 27x - 18$$

because

$$\begin{aligned} f(2) &= 2(2)^4 + 7(2)^3 - 4(2)^2 - 27(2) - 18 = 0 \\ f(-3) &= 2(-3)^4 + 7(-3)^3 - 4(-3)^2 - 27(-3) - 18 = 0. \end{aligned}$$

2. $x - 1$ is a factor of $f(x) = x^2 - 3x + 2$ because

$$f(1) = 1^2 - 3(1) + 2 = 1 - 3 + 2 = 0.$$

3. $x + 2$ is a factor of $f(x) = x^3 + 8$ because

$$f(-2) = (-2)^3 + 8 = -8 + 8 = 0.$$

REAL ZEROS OF A POLYNOMIAL FUNCTION

If $f(x)$ is a polynomial function and a is a real number, then the following statements are equivalent.

1. $x = a$ is a zero of the function f .
2. $x = a$ is a solution of the polynomial equation $f(x) = 0$.
3. $(x - a)$ is a factor of the polynomial $f(x)$.
4. $(a, 0)$ is an x -intercept of the graph of f .

Finding zeros of polynomial functions is closely related to factoring and finding x -intercepts.

Example 9. *Finding Zeros of a Polynomial Function*

1. Find all real zeros of

$$f(x) = x^3 - x^2 - 2x.$$

Solution.

$$f(x) = x^3 - x^2 - 2x$$

$$0 = x^3 - x^2 - 2x$$

$$0 = x(x^2 - x - 2)$$

$$0 = x(x - 2)(x + 1)$$

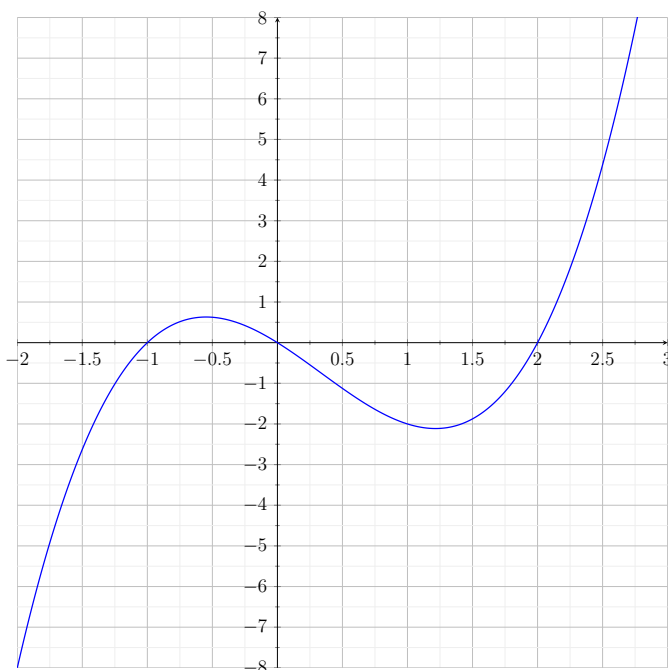
So, the real zeros are

$$x = 0, \quad x = 2 \quad \text{and} \quad x = -1,$$

and the corresponding x -intercepts are

$$(-1, 0), \quad (0, 0), \quad \text{and} \quad (2, 0).$$

Graph of $f(x) = x^3 - x^2 - 2x$.



2. Find all real zeros of

$$f(x) = -2x^4 + 2x^2.$$

Solution.

$$f(x) = -2x^4 + 2x^2$$

$$0 = -2x^4 + 2x^2$$

$$0 = -2x^2(x^2 - 1)$$

$$0 = -2x^2(x - 1)(x + 1)$$

So, the real zeros are

$$x = 0, \quad x = 1 \quad \text{and} \quad x = -1,$$

and the corresponding x -intercepts are

$$(0, 0), \quad (1, 0) \quad \text{and} \quad (-1, 0).$$

Graph of $f(x) = -2x^4 + 2x^2$.

